

Benchmark 2 Study Guide

PSYC 4213 Statistics · Modules 4-5 (Chapters 4-6)

Graded Benchmark 2 is proctored in Canvas (Respondus LockDown Browser + Webcam). No aids on the real exam. This guide matches the LMS Benchmark 2 study topics.

What this benchmark covers

- Module 4 - Descriptive statistics (Chapter 4)
- Module 5 - Graphing & visualization (Chapters 5-6 pragmatic matters)
- Central tendency, variability, distribution shape, standard scores, and choosing graphs.
- Jamovi (or your software) applications for descriptives and basic graphs.

Module 4 - Descriptive statistics

- Mean, median, mode - when each is appropriate; effect of outliers and skew.
- Range, IQR, variance, standard deviation; sum of squared deviations (SS).
- Z-scores: formula $Z = (X - \text{mean}) / \text{SD}$; converting z to raw scores ($X = \text{mean} + z \times \text{SD}$).
- Skewness and kurtosis; normal vs non-normal distributions (intro).
- Grouped descriptives; comparing summaries across groups.
- Levels of measurement: which descriptives/graphs are valid for nominal vs ordinal vs continuous data.

Module 5 - Graphing & visualization

- Histograms for continuous distributions; reading shape, center, spread.
- Box plots: median, IQR, outliers, comparing groups.
- Bar charts / pie charts for categorical counts (not for continuous means without context).
- Scatterplots for two continuous variables; direction and strength of association (visual).
- Frequency tables and contingency tables.
- Filtering data, logical expressions, transforming variables (compute, recode, log, etc.).
- Choose the right graph: categorical vs continuous; one vs two variables.

Common exam pitfalls

- Using a bar chart of means when a histogram of the raw distribution is needed (or vice versa).
- Reporting the mean for ordinal or highly skewed data without justification.
- Confusing SD with range; forgetting units when interpreting z-scores.
- Mislabeling axes or measurement level when importing data into software.

How to study

- Review Methods Market Topics: central tendency, variability, skew/kurtosis, standard scores, histograms, boxplots, bar charts, scatterplots.
- Complete Module 4-5 Concept Review and Software Practice.
- Practice Benchmark 2 formative test in Methods Market (30 questions per attempt).
- Work Module 4 Descriptive Statistics assignment-style problems (Compute formulas in Jamovi).

Methods Market: <https://methods-market.cneu.com/class/statistics> - Concept Review, practice tests, and Assignment Help for each module.